

HOW TO USE THIS GLOSSARY

This glossary defines the standard terminology used across all NuBred documents, the platform UI, the database schema, and the AI prompts. Every term listed here has a precise meaning in the NuBred context that may differ from general agricultural or legal usage.

Three rules apply to everyone working on NuBred:

- Use these terms consistently. Do not substitute synonyms — for example, do not use 'variety' when you mean 'selection', or 'forecast' when you mean 'harvest actual'.
- When in doubt, check here first. If a term you need is not in this glossary, flag it before inventing a definition.
- NuBred is flexible. If a project introduces a term not listed here, NuBred defines it in context and adds it to the project's local glossary. This does not change the standard glossary — it extends it for that project only.

Important: the NuBred Analysis Engine uses this glossary as one of its three analysis layers. Claude searches uploaded documents for concepts mappable to these terms — even when the source documents use different language. A contract written in Italian or Spanish will be mapped to the English NuBred standard terms defined here.

1 · GENETIC MATERIAL

Term	NuBred Definition
Genetic Material	Any plant material carrying heritable traits — seeds, cuttings, buds, tissue, pollen, or whole plants. The broadest category in NuBred.
Genotype	A specific genetic identity — any individual plant or group of plants sharing the same genetic makeup, whether or not it is protected by intellectual property rights.
Selection	A genotype that has been identified as potentially valuable through breeding or observation but has not yet been granted plant variety protection. A Selection may become a Variety if it passes the required phases and receives IP protection.
Variety	A genotype that has been granted plant variety protection (PBR, CPVR, or equivalent national right) and has a registered denomination. In NuBred, Variety always implies protected status.
Essentially Derived Variety (EDV)	A variety that is predominantly derived from another protected variety (the initial variety) and is essentially similar to it. Under UPOV, the holder of the initial variety retains rights over EDVs. EDVs discovered during a NuBred project must be reported immediately and assigned per contract terms.
Propagating Material	Any plant material used to reproduce a variety — including seeds, cuttings, runners, buds, grafts, tissue culture, and mother plants. The specific forms authorized for each project are defined in the contract and tracked in NuBred.

Mother Plant	A plant retained specifically to produce propagating material. Mother Plants represent the upstream control point of the germplasm chain. NuBred tracks their authorized existence and destruction per contract obligations.
Rootstock	A plant whose root system is used as the base onto which a scion (the desired variety) is grafted. Common rootstocks in kiwi: Hayward, Tomuri. NuBred records the rootstock used per planting unit.

2 · PROJECT STRUCTURE & ACTORS

Term	NuBred Definition
Project	The primary organizational unit in NuBred. A Project represents the full lifecycle of one or more varieties in a defined geographic ecosystem — from initial trials through commercial scale. Every actor, contract, phase, observation, and obligation in NuBred belongs to a Project.
Sub-Project	A Project nested within a parent Project. Used when a variety is tested or developed in multiple distinct locations or under distinct agreements that share a common origin but operate independently.
Ecosystem	The full network of actors, relationships, contracts, and commercial flows that surround a variety. NuBred represents the Ecosystem as an interactive diagram.
Variety Manager (VM)	The actor responsible for governing the project ecosystem. Typically a breeder, licensor, or Independent Variety Manager (IVM). In NuBred, the VM is the only role that creates projects, invites actors, and makes phase decisions.
Independent Variety Manager (IVM)	A company or individual with a mandate from the Breeder to develop and commercialize a variety in a specific territory. The IVM manages the ecosystem on behalf of the Breeder and is typically responsible for royalty collection and distribution.
Breeder	The entity that created or owns the intellectual property rights to a variety. May also act as Variety Manager, or may delegate this role to an IVM.
Licensed Grower	A producer authorized by the Variety Manager to plant and harvest a specific variety under the terms of a license agreement. Each Licensed Grower has defined planting rights, reporting obligations, and royalty obligations.
Nursery	A facility authorized to propagate and supply licensed plant material. Nurseries operate under a propagation license and are responsible for ensuring the genetic integrity and phytosanitary quality of the material they produce.
Packhouse	A facility that receives harvested fruit from Licensed Growers, sorts and grades it, and prepares it for commercial sale. In NuBred, the Packhouse provides aggregated production data used in royalty calculations.
Marketer	The commercial entity responsible for selling the fruit to retail or wholesale channels. May be the same as the Packhouse or a separate entity. Defines the commercial price used in delta EBITDA calculations.
Retailer	The end-of-chain commercial actor selling fruit to the consumer. Retailers require consistent supply and controlled positioning — NuBred governance directly supports these requirements.

3 · DEVELOPMENT PHASES

Term	NuBred Definition
Trial	The first development phase. Objective: screen multiple Selections with few plants per Selection to identify which have commercial potential. No commercial activity is permitted. Observations follow the experimental protocol defined for that species. Gate to advance: all protocol observations completed + VM decision per genotype (Promote / Repeat / Discard).
Pilot	The second development phase. Objective: validate a small number of Selections at semi-commercial scale — typically 0.5 to 1 hectare. Observations continue plus the production forecast cycle begins. The Quality Specification must be defined before Pilot can close. Gate to advance: observations completed + production forecast validated + Quality Specification defined + VM confirmation.
Launch	The first commercial phase. The variety enters the market under the governance structure defined in the contract. PBR and Brand royalties become active. Production forecasts and harvest actuals are mandatory. The Quality Specification governs which fruit qualifies for brand royalty.
Scale	The expansion commercial phase. The variety grows according to the planting plan defined in the contract. Delta EBITDA calculation is active in the Advanced Protocol. All commercial obligations and royalty flows are fully operational.
Phase Gate	The set of conditions that must be met before a project can advance from one phase to the next. NuBred checks gate conditions automatically and notifies the VM. The VM must confirm the advance decision. Phase Gates cannot be bypassed.
Promote	The VM's decision to advance a specific genotype from the current phase to the next. Recorded in NuBred with timestamp, source data, and VM identity.
Repeat	The VM's decision to continue the current phase for a specific genotype because more data is needed. A new observation cycle opens automatically.
Discard	The VM's decision to remove a specific genotype from active development. The genotype is archived with its full observation history — it is never deleted. Archived genotypes can be reinstated if new evidence warrants it.

4 · CONTRACTS & LEGAL INSTRUMENTS

Term	NuBred Definition
Plant Variety Protection (PVP / PBR)	The intellectual property right protecting a plant variety. Grants the holder exclusive rights over propagation, production, and commercialization of the variety. In the EU, this is administered by CPVO (Community Plant Variety Office). NuBred tracks PBR status, filing dates, expiry dates, and territories.
CPVR	Community Plant Variety Right — the EU-level plant variety protection administered by CPVO. Grants rights across all EU member states from a single application.
UPOV	International Union for the Protection of New Varieties of Plants — the intergovernmental organization that establishes the framework for plant variety protection globally. NuBred uses UPOV descriptors as the standard for qualifying genotype characteristics.

VCU	Value for Cultivation and Use — the formal assessment of whether a variety has sufficient agronomic and commercial merit to justify protection and commercialization. Trial phase activities are typically organized around VCU evaluation.
License Agreement	A contract granting a Licensee specific rights over a protected variety — typically propagation, cultivation, and/or commercialization rights — in exchange for royalty payments. NuBred analyzes License Agreements to extract project structure.
Exclusive License	A license where the Licensor grants rights to a single Licensee in a defined territory, committing not to license the same rights to any other party in that territory. Exclusivity conditions and maintenance thresholds are tracked in NuBred.
Sub-License	A license granted by a Licensee to a third party, subject to the Licensor's prior written authorization. NuBred tracks the sub-license chain and its obligations.
Trials Agreement	A contract granting a Cooperator the right to evaluate a variety under strictly defined conditions — no commercial activity, no propagation, no publication without consent. All trial data belongs to the Breeder. This is the contract type governing the Trial phase.
VMC Agreement	Variety Management Company Agreement — the contract under which a Breeder delegates the commercial development of a variety to an IVM. Defines the phase structure, option rights, royalty flow, and mandate scope.
Quality Specification	The document defining the minimum quality standards a fruit must meet to qualify for brand royalty payment. Defined by the end of the Pilot phase as a mandatory gate condition. Covers parameters such as minimum Brix, minimum calibre, colour requirements, and pack-out percentage.
Planting Plan	The structured plan defining how many hectares of a variety will be planted, by whom, in which locations, and on what timeline. Planting Plans are extracted from contracts and tracked against actual planting declarations in NuBred.
Minimum Quantities	The minimum volume of propagating material or planted hectares a Licensee must achieve to maintain exclusivity or meet contract obligations. Non-compliance may trigger exclusivity loss. Tracked in NuBred against declared figures.
Entry Fee	A one-time payment made by the Licensee at contract signing as consideration for acquiring exclusive rights. May be partially refundable if PBR is rejected. Recorded in NuBred as part of the financial structure of the project.

5 · COMMERCIAL TERMS & ROYALTIES

Term	NuBred Definition
Royalty	A periodic payment made by a Licensee or Licensed Grower to the rights holder in exchange for the authorized use of a protected variety. Royalties may be calculated per plant, per hectare, per kilogram of fruit, or as a percentage of commercial value.
PBR Royalty	A royalty based on the plant variety protection right — typically charged per plant propagated or per hectare planted. Collected by the Breeder or IVM from Nurseries and Licensed Growers.

Brand Royalty	A royalty based on the commercial use of the variety's brand — typically calculated as a percentage of the exwork or FOB price per kilogram of fruit that meets the Quality Specification. Collected by the IVM from Packhouses or Marketers.
Exwork Price	The price of fruit at the point it leaves the producer's facility, before transport and distribution costs. Used as the basis for Brand Royalty calculation in many agreements.
FOB Price	Free on Board — the price of fruit at the point it is loaded onto the transport vessel or vehicle. Used as an alternative basis for Brand Royalty calculation in international agreements.
Production Forecast	A structured estimate of the production volume (kg/ha or total kg) expected from a given planting in a given season. In NuBred, the forecast follows a 4-measurement cycle: Fruit Set Count, Growing Stage Diameter, Ripening Stage Weight, and Harvest Actual. Mandatory from the Pilot phase onward.
Harvest Actual	The definitive production figure declared by the Licensed Grower after harvest is complete. Must be submitted within 30 days of harvest completion. This figure triggers the royalty calculation.
Benchmark Variety	The standard commercial variety used as a reference point for delta EBITDA calculation in the Advanced Protocol. Defined per species and market profile in the NuBred Benchmark Registry. The VM can select a different benchmark for a specific project but must document the reason.
Delta EBITDA	The difference between the EBITDA generated by the protected variety and the EBITDA that would have been generated by the Benchmark Variety under the same conditions. In the Advanced Protocol, royalties are calculated as a percentage of the Delta EBITDA — aligning compensation with real economic value creation.
Benchmark Registry	The NuBred-maintained database of benchmark varieties per species and market profile. Used as the reference for delta EBITDA calculations. Variety Managers can read the registry but cannot edit it. Maintained by NuBred administrators.
Pack-out Percentage	The ratio of commercially graded fruit to total harvested weight. A key indicator of variety performance and a component of both Quality Specification compliance and EBITDA calculation.
Withholding Tax	A tax deducted at source on royalty payments flowing across borders. Relevant in VMC Agreement structures where an IVM collects royalties locally and remits net amounts to the Breeder abroad. NuBred tracks withholding tax obligations where applicable.

6 · OBSERVATIONS & PROTOCOLS

Term	NuBred Definition
Observation	A structured data collection event performed at a defined point in the project lifecycle. Observations are organized by Parameter Family, follow the protocol defined for the current phase, and are recorded against specific genotypes or planting units.
Parameter Family	One of seven fixed categories that organize all observation parameters in NuBred: Plant Characteristics (F1), Fruit Quality (F2), Production (F3),

	Phytosanitary (F4), Post-Harvest (F5), Climate (F6), Commercial (F7). Families are fixed and never change across species or projects.
Protocol	The set of parameters, measurement methods, frequencies, and timing windows that govern data collection for a specific phase of a project. Protocols are extracted from uploaded documents and matched against NuBred standard templates.
Threshold	The minimum or target value a parameter must reach for a genotype to meet phase gate or Quality Specification requirements. NuBred recognizes four threshold types: Quantitative Absolute, Quantitative Relative, UPOV Qualitative, and Boolean.
Eliminatory Criterion	A threshold whose breach automatically triggers a Discard signal for the genotype, regardless of performance on other parameters. NuBred defines three levels: L1 Universal (fixed by NuBred, cannot be overridden), L2 Species Standard (NuBred proposes, VM can raise but not lower), L3 Custom (VM-defined for project-specific requirements).
Weighted Score	A composite performance score calculated from non-eliminatory parameters, each assigned a weight within a species-and-market-specific profile. Used to compare genotypes and support phase gate decisions. Weights can be adjusted by the VM within defined ranges.
BBCH Scale	A standardized numerical scale for describing the phenological development stages of plants. Used in NuBred for phenology observations. Values range from 00 (dry seed) to 99 (secondary dormancy). NuBred uses BBCH as the standard phenology input type.
Brix	A measure of the sugar content of fruit juice, expressed as a percentage of dissolved solids. Measured with a refractometer. One of the primary quality parameters in NuBred and a key Eliminatory Criterion for most fruit species.
VCU Assessment	The formal evaluation of a variety's Value for Cultivation and Use — conducted during the Trial phase. Results are owned exclusively by the Breeder and cannot be published without written authorization. NuBred tracks VCU assessment completion as part of Trial phase gate logic.
Post-Harvest Evaluation	An observation conducted on stored fruit after harvest to assess storage duration, disease development, weight loss, and physical defects (pitting, pebbling). Mandatory in the Pilot phase. Typically evaluated at 15 and 30 days post-harvest.

7 · PHYTOSANITARY & QUARANTINE

Term	NuBred Definition
Quarantine	A legally mandated period during which imported plant material is isolated and inspected before being released for use. Duration and conditions are defined by the receiving country's phytosanitary authority. NuBred tracks quarantine status, expected certification dates, and compliance documentation.
Phytosanitary Certificate	An official document issued by the national plant protection authority certifying that the plant material meets the importing country's phytosanitary requirements. Required for international movement of propagating material.
Cracking	A physiological disorder in fruit where the skin splits due to rapid water absorption — common in cherries and other thin-skinned varieties. In NuBred,

	cracking exceeding 30% of sampled fruit is an Eliminary Criterion L2 (species standard) for cherry.
Monilia	Monilinia spp. — a fungal pathogen causing brown rot in stone fruits. High susceptibility is an L2 Eliminary Criterion in cherry. Affection above 20% at harvest triggers a Discard signal.
Psa	Pseudomonas syringae pv. actinidiae — a bacterial pathogen affecting kiwifruit. High susceptibility is monitored as a phytosanitary parameter in kiwi protocols.

8 · AI & SYSTEM TERMS

Term	NuBred Definition
NuBred Analysis Engine	The AI-powered component of NuBred that reads uploaded documents and generates the structured project representation. Built on Claude API (Anthropic) for contract analysis and deep reasoning, and Gemini API (Google) for lighter tasks such as observation data assistance.
Project Representation	The structured output generated by the NuBred Analysis Engine after processing uploaded documents. Includes the Actor Diagram, Phase Timeline, Clause Analysis, and To-Do List. The Project Representation is a draft until confirmed by the VM.
Actor Diagram	The visual representation of all actors in a project and their relationships. Actors appear as nodes; relationships appear as labelled edges showing the contract that governs each connection. Every element is traceable to a source document.
Source Citation	The reference attached to every conclusion generated by the NuBred Analysis Engine — showing the document name, relevant passage, and the NuBred framework element (glossary term, phase definition, or clause) that guided the mapping. Ensures every AI conclusion is verifiable.
To-Do List	The list of obligations for each actor in a project, organized by season and type. Generated from two sources: the NuBred standard framework for that phase (structural obligations) and the specific contracts uploaded for the project (contractual obligations). Updated dynamically as the project progresses.
Review Screen	The three-tab interface shown to the VM after AI analysis is complete. Tab 1: Actor Diagram. Tab 2: Clause Analysis (HIGH and MEDIUM risk clauses + all absent clauses). Tab 3: Gaps (all ABSENCE, ANOMALY, and IMBALANCE flags). The VM must view Tab 3 before confirming the project.
Draft Project	A project that has been analyzed by the AI but not yet confirmed by the VM. Draft Projects exist in NuBred but are not visible to invited actors. The VM can edit a Draft before confirming.
Quality Specification Gate	The phase gate condition requiring that the Quality Specification document be defined before the project can advance from Pilot to Launch. This gate cannot be bypassed.

9 · FLEXIBLE & PROJECT-SPECIFIC TERMS

NuBred is not a rigid system. The following terms define how NuBred handles concepts that go beyond the standard framework. This flexibility is a deliberate design choice — the system must be able to describe any varietal project accurately, even when that project uses non-standard terminology or structures.

Term	NuBred Definition
Project-Specific Term	A term identified by the NuBred Analysis Engine in uploaded documents that does not exist in the standard NuBred Glossary but is meaningful for the specific project. NuBred defines the term in context and adds it to the project's local glossary. Project-specific terms are visible only within that project.
Project-Specific Clause	A contract clause identified by the NuBred Analysis Engine that does not map to any of the 45 standard AgreeLyze clauses but is material to understanding the project structure. NuBred describes the clause, flags its risk direction, and includes it in the project's clause record.
Custom Phase	A development phase with a name, duration, gate criteria, and protocol that differs from the four NuBred standard phases. Available on request for projects with complex or non-standard structures. Custom Phases are configured by the NuBred team after reviewing the project's contracts — they are not self-service.

GLOSSARY MAINTENANCE

This glossary is a living document. It will be updated as NuBred evolves and as new project types are onboarded.

- Standard terms (Sections 1–8) are maintained by the NuBred product owner. Changes require a version update to this document and to all affected prompts and database schemas.
- Project-specific terms (Section 9) are added automatically by the NuBred Analysis Engine during project setup. They are stored in the project record and do not affect the standard glossary.
- If a project-specific term appears repeatedly across multiple projects, the product owner may decide to promote it to the standard glossary in the next version.

Current version: v1 — April 2026. Next review: when a new species is added to the platform or when a new contract family is onboarded.